Exam Date & Time: 19-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CSE STREAM) EXAMINATIONS - NOVEMBER 2024
SUBJECT: MAT 1172 - COMPUTATIONAL MATHEMATICS - I
COMPUTATIONAL MATHEMATICS - I [MAT 1172]

Marks: 50 **Duration: 180 mins.**

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Answer all the questions.

1A) Find all the eigenvalues and any two eigenvectors of the matrix $A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ (4)

1B) Test the consistency and hence solve by Gauss elimination method $2x + y + 4z = 12,\ 4x + 11y - z = 33, 8x - 3y + 2z = 20.$ (3)

1C) Use Gauss-Seidel method to find an approximate solution for the system of linear equations $2x + y + 6z = 9, 8x + 3y + 2z = 13, x + 5y + z = 7$, correct to four decimal places. Take the initial approximation as $x^{(0)} = y^{(0)} = z^{(0)} = 0$, carry out four iterations. (3)

2A) Use Gram-Schmidt orthogonalization process to construct an orthonormal basis of $\mathbb{R}^3$ from the given set of vectors $\{(2,3,0),\ (6,1,0), (0,2,4)\}$. (4)

2B) Test whether the set $\{(5,-2,4), (2,-3,5), (-4,5,-7)\}$ forms a basis for $\mathbb{R}^3$ or not. If so, express $(-1,2,1)$ in terms of basis vectors. (3)

2C) Let $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$ be the matrix representation of a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^3$ with respect to the basis $\{(1,1), (1,-1)\}$ and $\{(1,1,0), (1,-1,0), (0,0,1)\}$ of $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively. Find a formula for T. (3)

3A) Solve the differential equation $2\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y = x^2 + 2x - 1$ (4)

3B) Use the method of variation of parameters to solve

$$y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$$ (3)

3C) Trace the curve $x = a(t - sint),\ y = a(1 - cost)$ where $a > 0$ and $0 < t < 2\pi$. (3)

4A) Diagonalize the matrix $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ and hence find $Trace(A^5)$. (4)

4B) (3)

Use Regulai- Falsi method to find a real root of the equation $4e^{-x}sinx - 1 = 0$ in $(0,0.5)$. Perform 3 iterations.

4C) Use Runge-Kutta method of order four to find $y(1.5)$, Given that $\frac{dy}{dx} = \frac{2x-y^2}{x^2+y}, y(1) = 0$. Take $h = 0.5$. (3)

5A) Use divided difference formula to find the value of $f(4)$ from the following data

x	0	2	3	6
$f(x)$	-4	2	14	158

(4)

5B) Find the value of cos (1.74) using the values given below

x	1.7	1.74	1.78	1.82	1.86
$y = \sin x$	0.9916	0.9857	0.9781	0.9691	0.9584

(3)

5C) Use Simpson's $\frac{1}{3}$ rule to evaluate $\int_1^5 \sqrt{1+x^2}\ dx$, by taking 4 sub-intervals. (3)

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